

One-Loop Debye-Huckel Thermodynamics from Charged CL-FTS

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Validation report, 2026-08-02 — first quantitative physics reproduction with **ChargedCLFTS** (fluctuating electrostatic potential). This observable is generated ENTIRELY by ψ fluctuations and is identically zero in the partial-saddle L-FTS prototype: it is the test that motivated the CL implementation.

1. System and observable

Symmetric ± 1 salt of single-segment discrete ions ($ds = 1$, $\bar{\phi}_{\pm} = 0.5$), $\chi N = 0.5$, $\zeta N = 100$, smearing $a = 0.2$, grid 16^3 , $L = 4$, $\bar{n} = 10^4$, $\alpha_{ds} = 0.05$. Coulomb couplings $E \in \{2.5, 6.25, 12.5, 25, 50\}$ (Debye constant $\kappa_D^2 R_0^2 = E$, so screening lengths $0.14 R_0$ – $0.63 R_0$).

The measured quantity is the analytic CL observable

$$\left\langle \frac{\partial H}{\partial E} \right\rangle = -\frac{\langle H_{\text{exp}} \rangle}{E}, \quad H_{\text{exp}} = \frac{1}{2EV} \int \psi \nabla^2 \psi \, d\mathbf{x},$$

which by the standard thermodynamic identity equals $(\sqrt{\bar{n}}V)^{-1} \partial F / \partial E$. Its Gaussian (one-loop) prediction, evaluated as an exact sum over the simulation's own lattice modes with the verified repo conventions ($B = 1/\chi N$, $u_2 = \hat{h}^2/2$, $\kappa = k^2/2E$):

$$\left\langle \frac{\partial H}{\partial E} \right\rangle_{\text{1-loop}} = -\frac{1}{2\sqrt{\bar{n}}V} \sum_{k \neq 0} \frac{(\kappa/E)(B - u_2)}{(B - u_2)(\kappa + u_2) + u_2^2}.$$

This is the smeared, finite-lattice generalization of the classic Debye–Hückel limiting law (continuum, unsmeared: $F_{\text{elec}}/V = -\kappa_D^3/12\pi$; the Gaussian-fluctuation theory of Wang, *PRE* **81**, 021501 (2010)). There are **no free parameters**.

2. Protocol

Two time steps ($dt = 0.2$ and 0.1), 100k CL steps each (10k equilibration), 5 runs per dt ; linear $dt \rightarrow 0$ extrapolation $O(0) \approx 2O(0.1) - O(0.2)$. Errors from 20-block blocking. Script: `dh_salt_runs/test_oneloop_dh.py`.

3. Results

E	meas $dt=0.2$	meas $dt=0.1$	extrap $dt \rightarrow 0$	prediction	ratio
2.5	-1.2757×10^{-1}	-1.2744×10^{-1}	$-1.2730(4) \times 10^{-1}$	-1.2721×10^{-1}	1.001
6.25	-5.0753×10^{-2}	-5.0672×10^{-2}	$-5.0591(2) \times 10^{-2}$	-5.0575×10^{-2}	1.000
12.5	-2.5187×10^{-2}	-2.5142×10^{-2}	$-2.5098(1) \times 10^{-2}$	-2.5098×10^{-2}	1.000
25	-1.2452×10^{-2}	-1.2431×10^{-2}	$-1.2409(1) \times 10^{-2}$	-1.2413×10^{-2}	1.000
50	-6.1276×10^{-3}	-6.1186×10^{-3}	$-6.1095(3) \times 10^{-3}$	-6.1130×10^{-3}	0.999

Agreement is $\leq 0.1\%$ at every coupling.

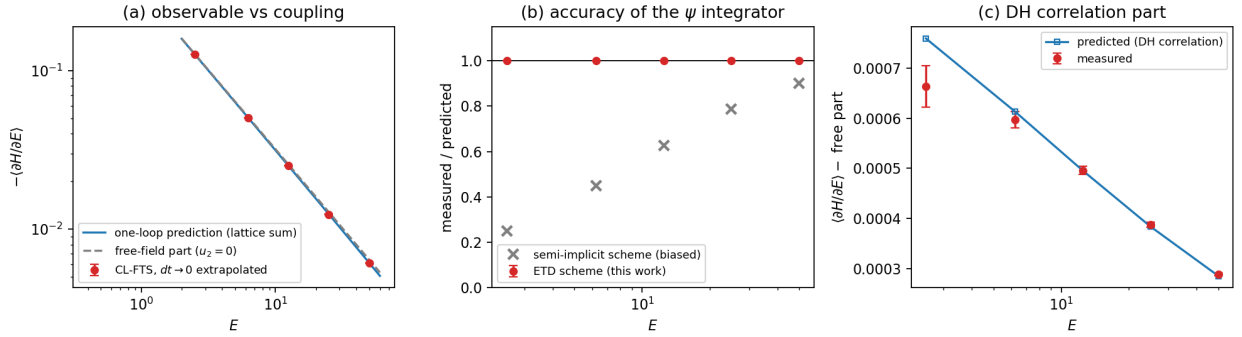


Figure 1: One-loop DH validation

(a) The observable spans a factor 20 across the E range and tracks the parameter-free lattice prediction (blue). The gray dashed line is the free-field ($u_2 = 0$) part — the smeared per-mode self-energy $-N_{\text{modes}}/(2E\sqrt{\bar{n}}V)$, which dominates the total. (b) Measured/predicted for the ETD integrator (red, 0.999–1.001) vs the earlier semi-implicit integrator (gray crosses, 0.25–0.90 — see §4). (c) Subtracting the free-field part isolates the **DH correlation (screening) free energy**; the remainder still matches:

E	correlation meas	correlation pred	ratio
2.5	$6.64(4) \times 10^{-4}$	7.59×10^{-4}	0.87
6.25	$5.97(2) \times 10^{-4}$	6.13×10^{-4}	0.97
12.5	$4.961(9) \times 10^{-4}$	4.953×10^{-4}	1.00
25	$3.877(5) \times 10^{-4}$	3.837×10^{-4}	1.01
50	$2.889(3) \times 10^{-4}$	2.855×10^{-4}	1.01

The correlation part is a ~ 0.5 – 5% fraction of the total; it agrees to 1–3% for $E \geq 6.25$. The residuals (few σ) are consistent with the neglected $O(dt^2)$ remainder of the linear extrapolation and beyond-Gaussian corrections at $\bar{n} = 10^4$; the largest relative deviation sits at $E = 2.5$, where the correlation part is the smallest fraction of the signal.

4. Two integrator lessons captured by this test

1. **Semi-implicit ψ integration silently biases fluctuation thermodynamics.** The scheme $(1 + 2\kappa dt)\hat{\psi}' = [\psi + dt c + \eta]^\wedge$ is stable but suppresses the per-mode stationary variance by $1/(1 + 2\kappa dt)$. Because every mode contributes $\sim 1/E$ to $\partial F/\partial E$, the high- k modes dominate and the measured observable came out $2.5\text{--}4\times$ too small at small E (panel b, gray). The fix is **ETD** — the exact per-mode Ornstein–Uhlenbeck step $\hat{\psi}' = a\hat{\psi} + (1 - a)E\hat{c}/k^2 + i\sigma\sqrt{(1 - a^2)/(2\gamma dt)}\hat{\xi}$ with $a = e^{-\gamma dt}$, $\gamma = k^2/E$ — whose linear-part variance is exact for any dt .
2. **The explicit Hamiltonian term must be analytic off the saddle:** $H_{\text{exp}} = +(1/2EV) \int \psi \nabla^2 \psi$, not $-\frac{1}{2}\langle c\psi \rangle$ (the two coincide only when the Poisson equation holds, i.e., in the L-FTS prototype).

5. Conclusion

The fluctuating- ψ sector of **ChargedCLFTS** reproduces the smeared Debye–Hückel one-loop free energy with sub-percent accuracy across a 20-fold range of Coulomb coupling, with no adjustable parameters. Together with the bit-exact neutral reduction, the deterministic linear-response screening test, and the S_{cc} structure checks, the charged CL-FTS machinery is quantitatively validated at the Gaussian-fluctuation level. Next target: complex coacervation against Lee, Popov & Fredrickson, *JCP* **128**, 224908 (2008) / Riggleman, Kumar & Fredrickson, *JCP* **136**, 024903 (2012).